Introduction to NLP

Lecture 2 - Phonetics

Agenda

- Speech Generation
- Speech Perception
- Computational Audio Processing
- The Spectrogram

Part 3



Part 1



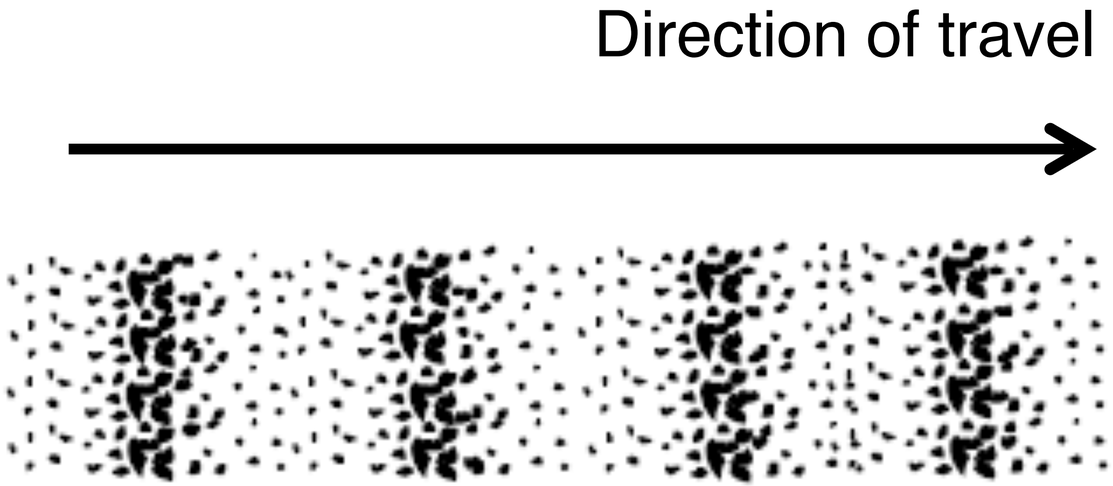
Part 2



What is sound?

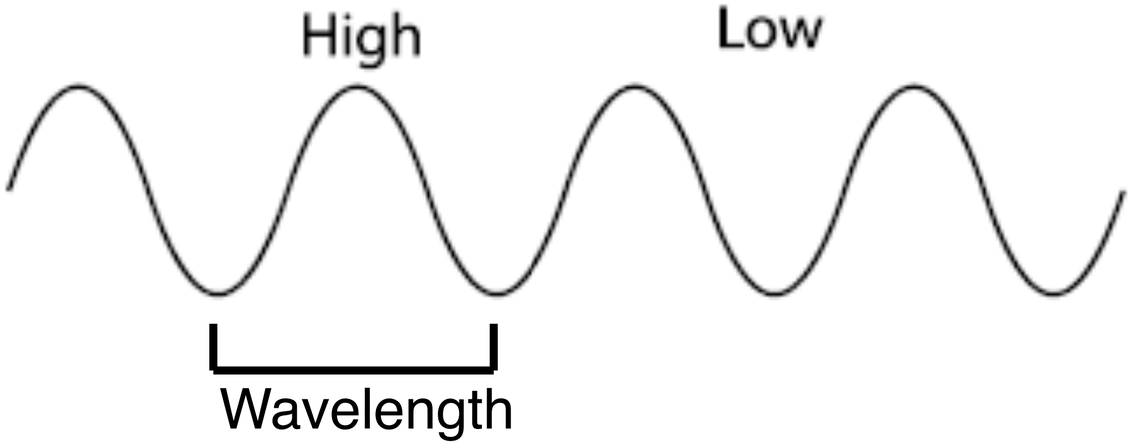
https://www.youtube.com/watch?v=Rbuhdo0AZDU

- Sound waves are *longitudinal waves*



- They travel as pressure differences through air, like this:

- But we usually represent them like this instead:



Sound and Language

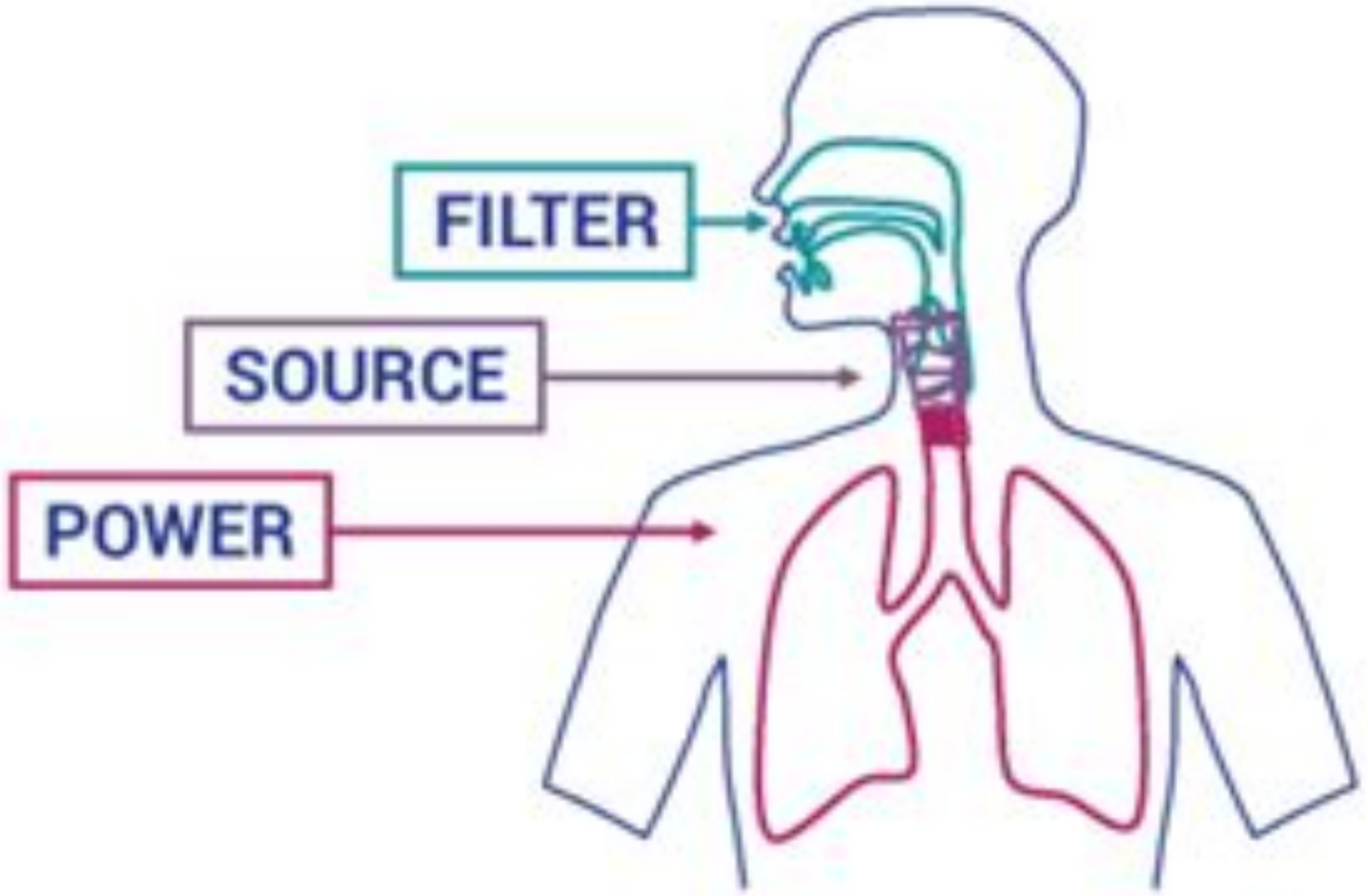
- Phone - smallest unit of sound (physical distinction)
- Phoneme - smallest unit to make a meaning change (language dependent)
- Morpheme - smallest meaningful unit

IPA

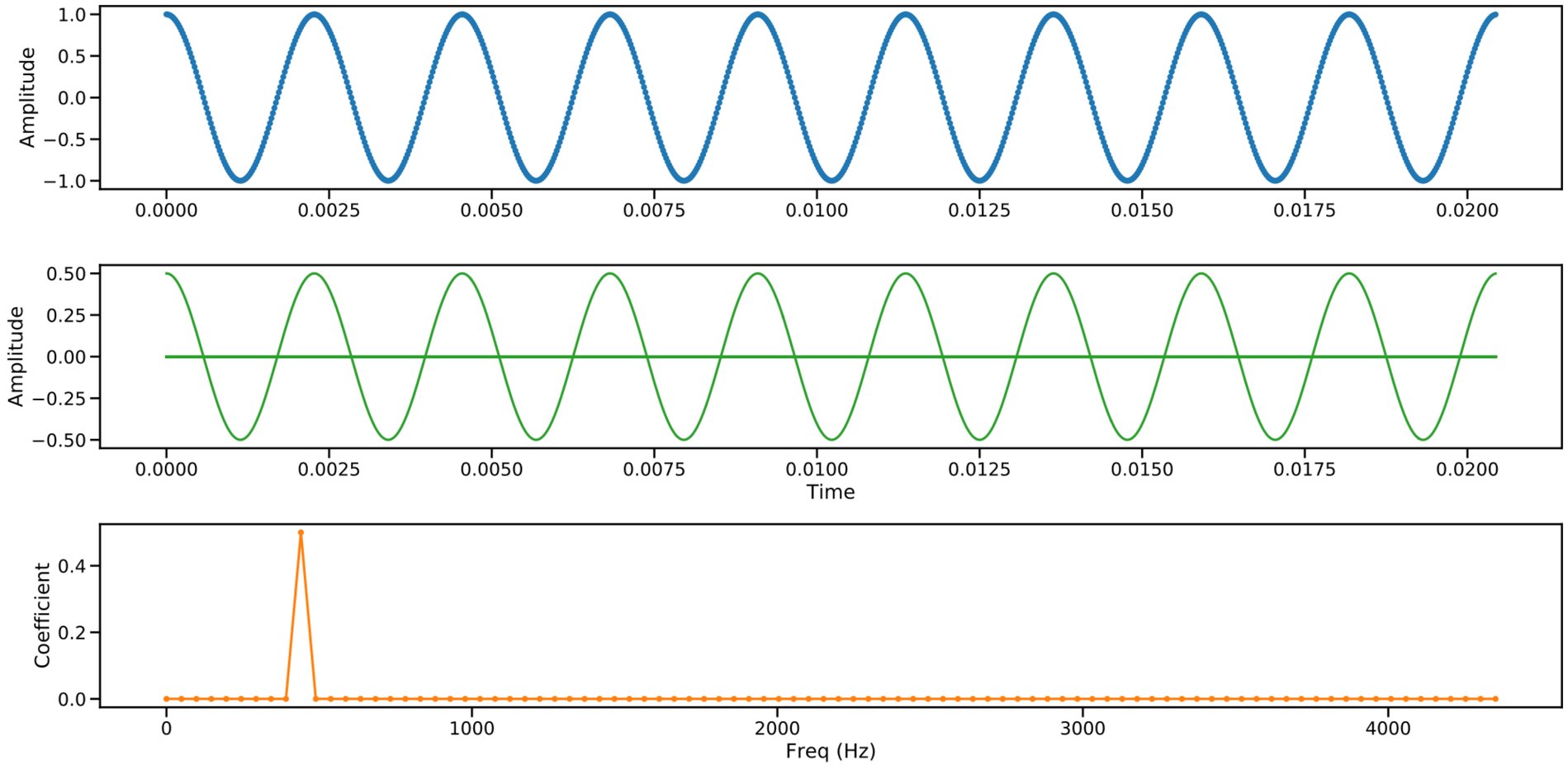
- International

[ɪntənæʃənəɫ]
[ɪĩ̃ɚ̃næʃɨnəɫ]

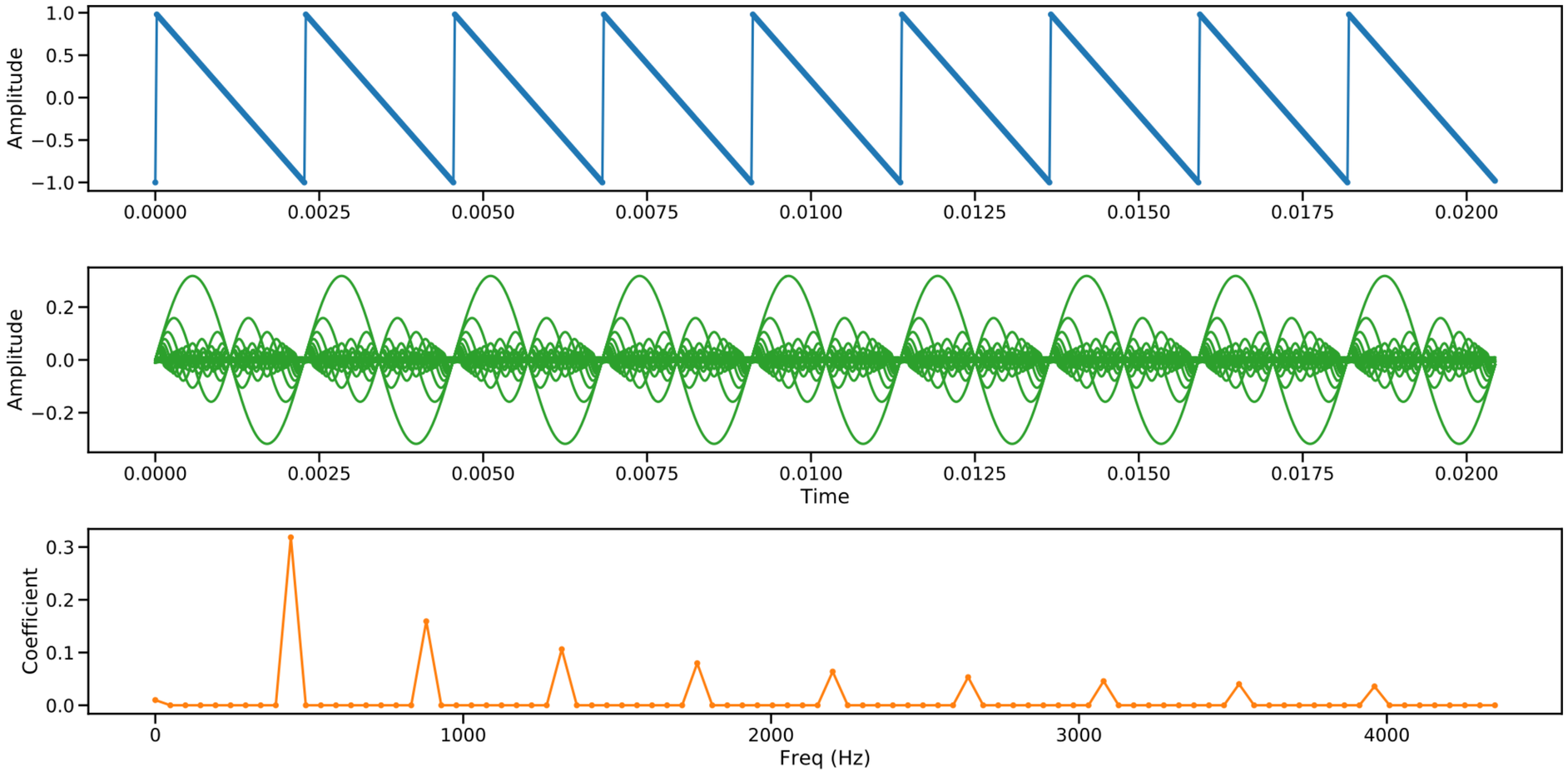
Language Production



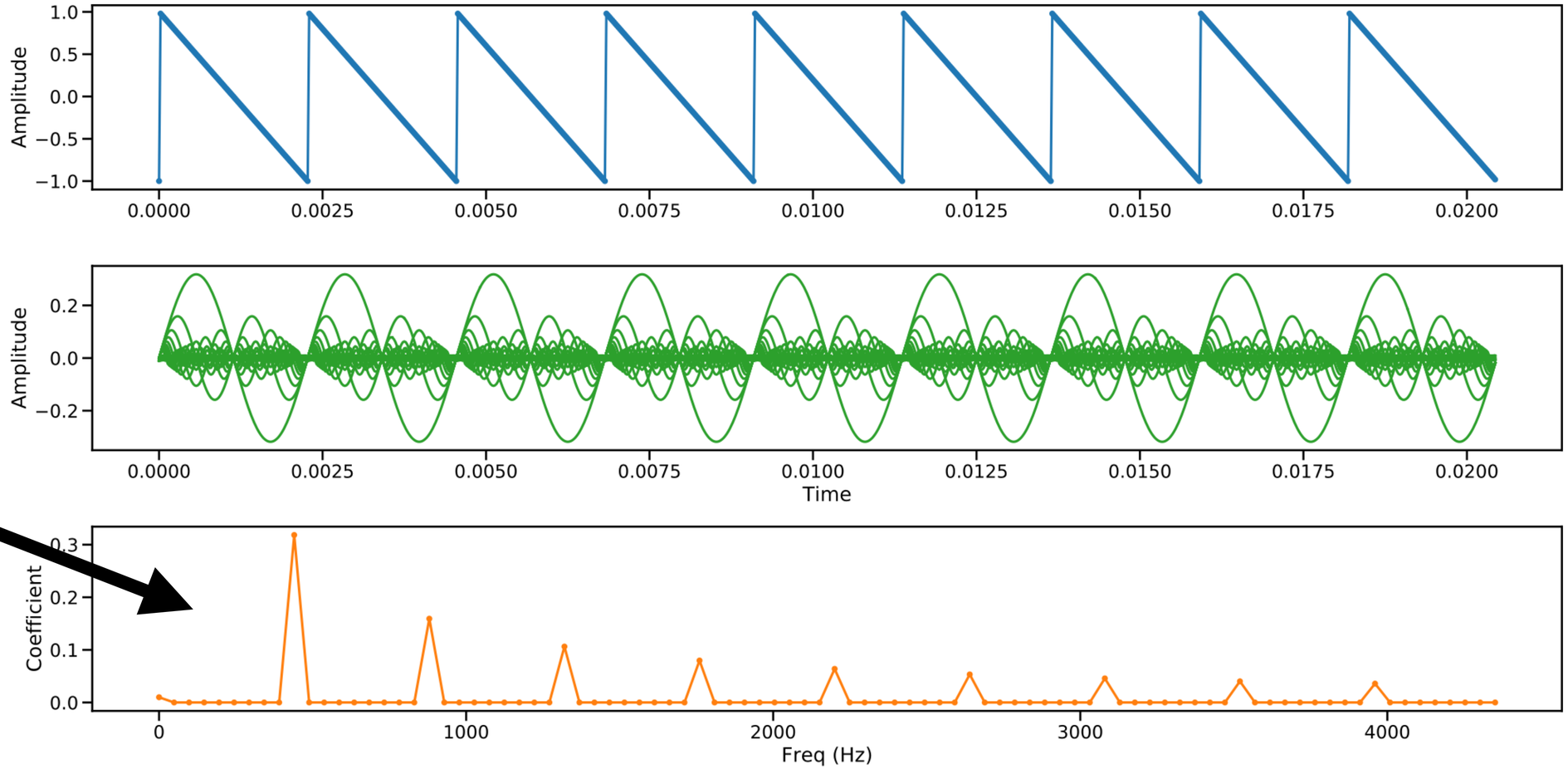
Sine Wave


Amplitude
1.0
0.5
0.0
−0.5
−1.0
0.0000
0.0025
0.0050
0.0075
0.0100
0.0125
0.0150
0.0175
0.0200
Amplitude
0.50
0.25
0.00
−0.25
−0.50
Time
Coefficient
0.4
0.2
0.0
0
1000
2000
3000
4000
Freq (Hz)

Sawtooth Wave

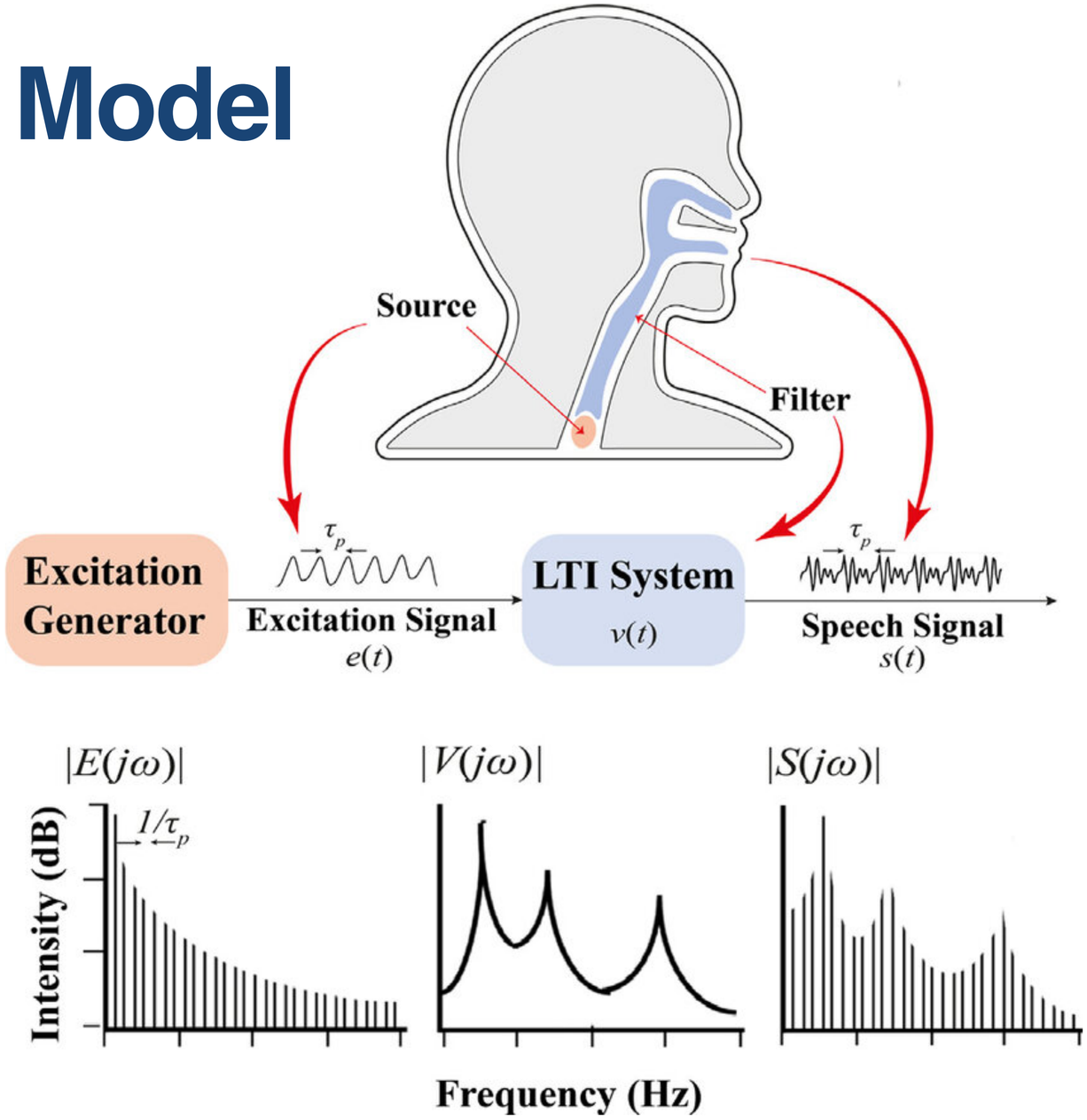


Sawtooth Wave



Harmonics

Source-Filter Model



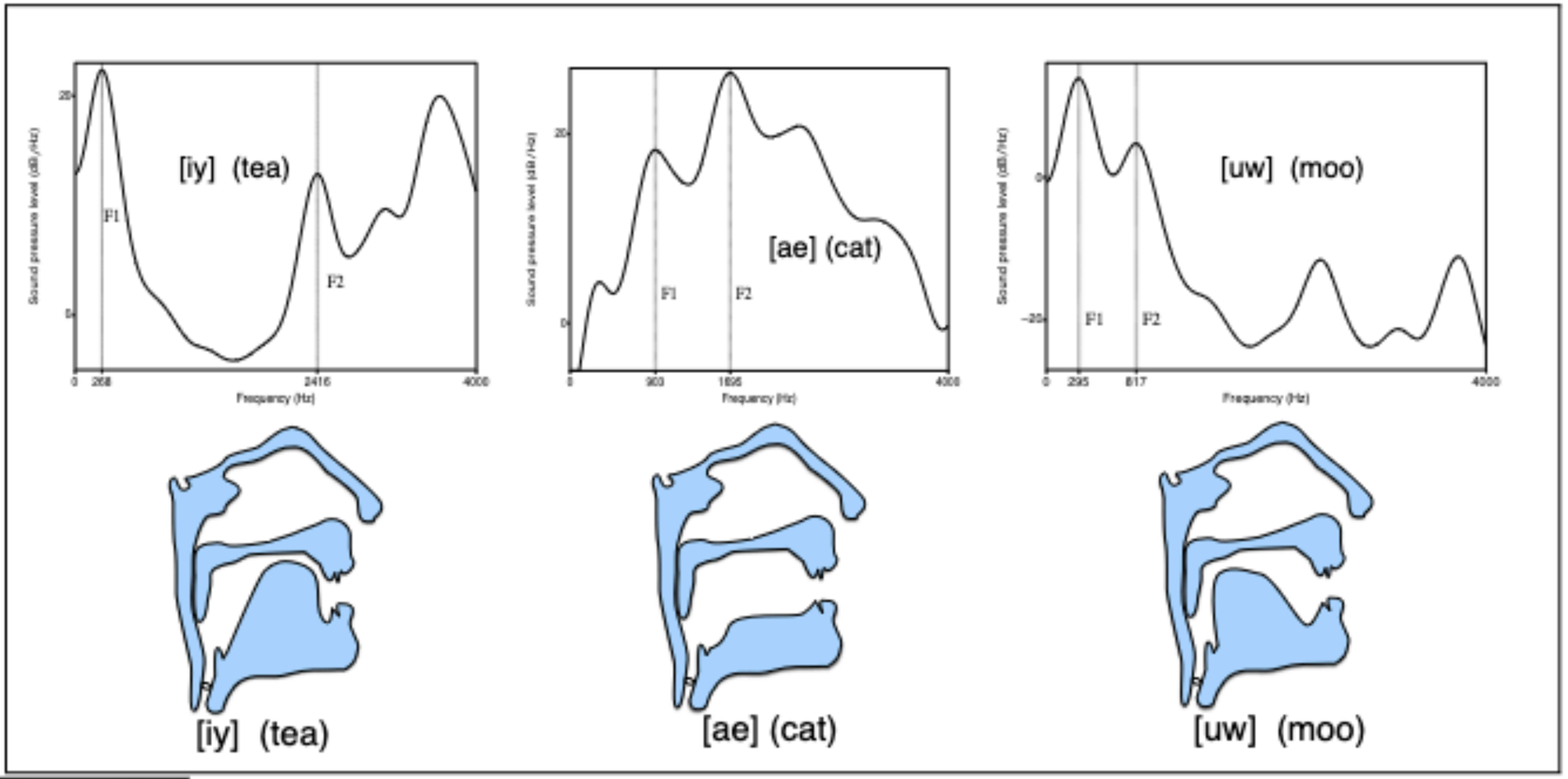
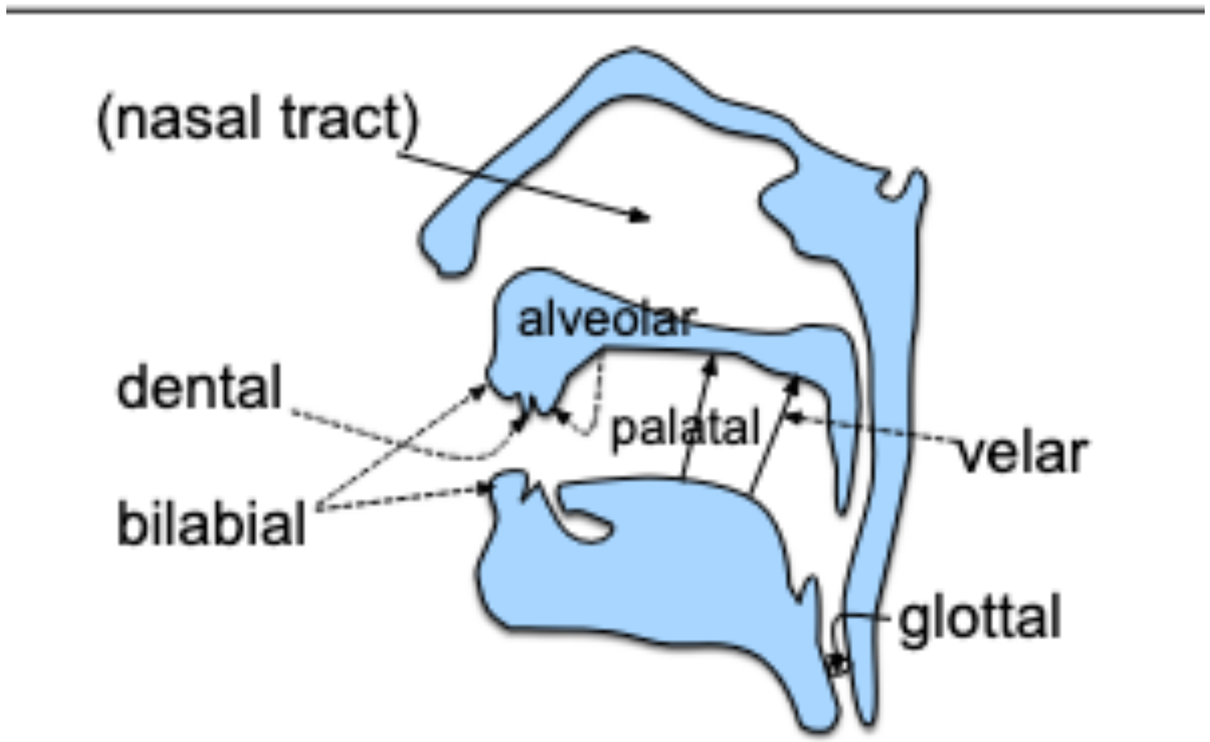


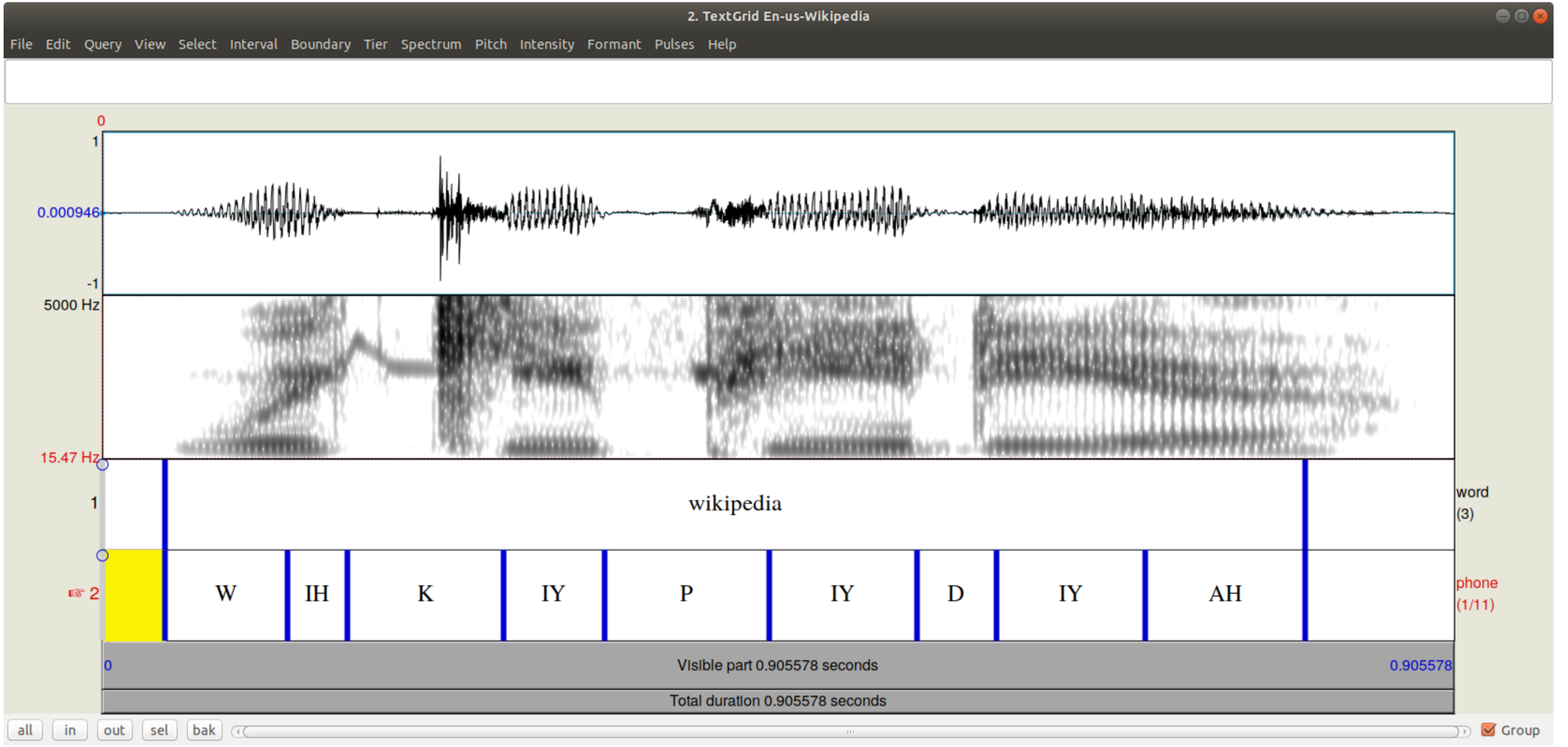
Figure 15.24 Visualizing the vocal tract position as a filter: the tongue positions for three English vowels and the resulting smoothed spectra showing F1 and F2.

Consonants

- Labial: [p]ossum, [b]ear, [m]armot
- Dental: [th]ing
- Alveolar: [s]nake, [z]ebra, [t]iger, [d]ragon
- Palatal: [sh]rimp, [ch]ina
- Velar: [g]oose, [k]rane
- Glottal: glottal stop
- Plosives: [p]
- Nasal: [n], [m]
- Fricatives: [f], [s]



https://commons.wikimedia.org/wiki/File:Real-time_MRI_-_Speaking_(English).ogv

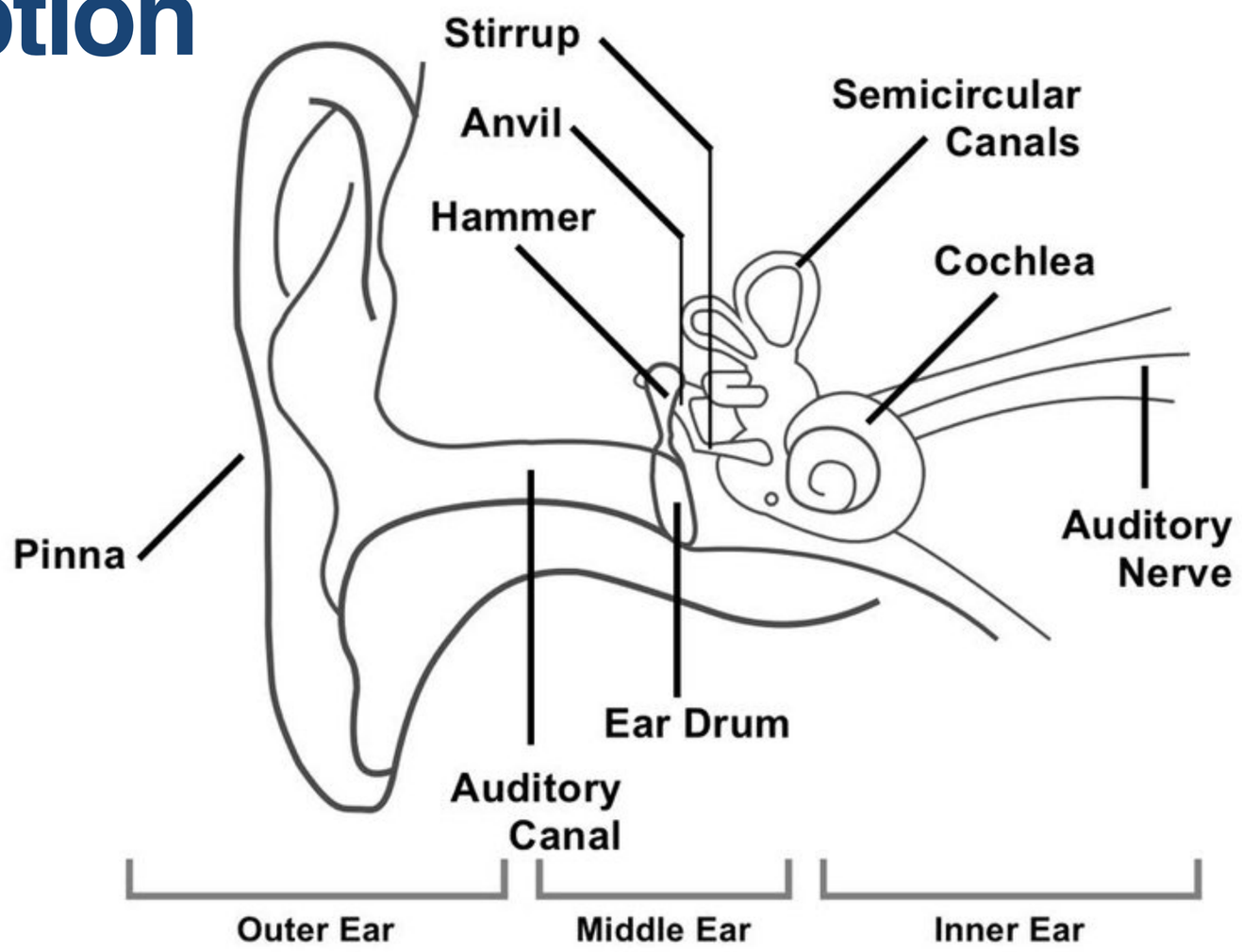

2. TextGrid En-us-Wikipedia
File Edit Query View Select Interval Boundary Tier Spectrum Pitch Intensity Formant Pulses Help
0
1
0.000946
-1
5000 Hz
15.47 Hz
1
wikipedia
word
(3)
2
W
IH
K
IY
P
IY
D
IY
AH
phone
(1/11)
0
Visible part 0.905578 seconds
0.905578
Total duration 0.905578 seconds
all
in
out
sel
bak
Group

Conclusion

- Speech is generated by modulating the sound waves coming from the vocal chords
- The shape of the whole vocal cavity determines what sound is produced

Language Perception

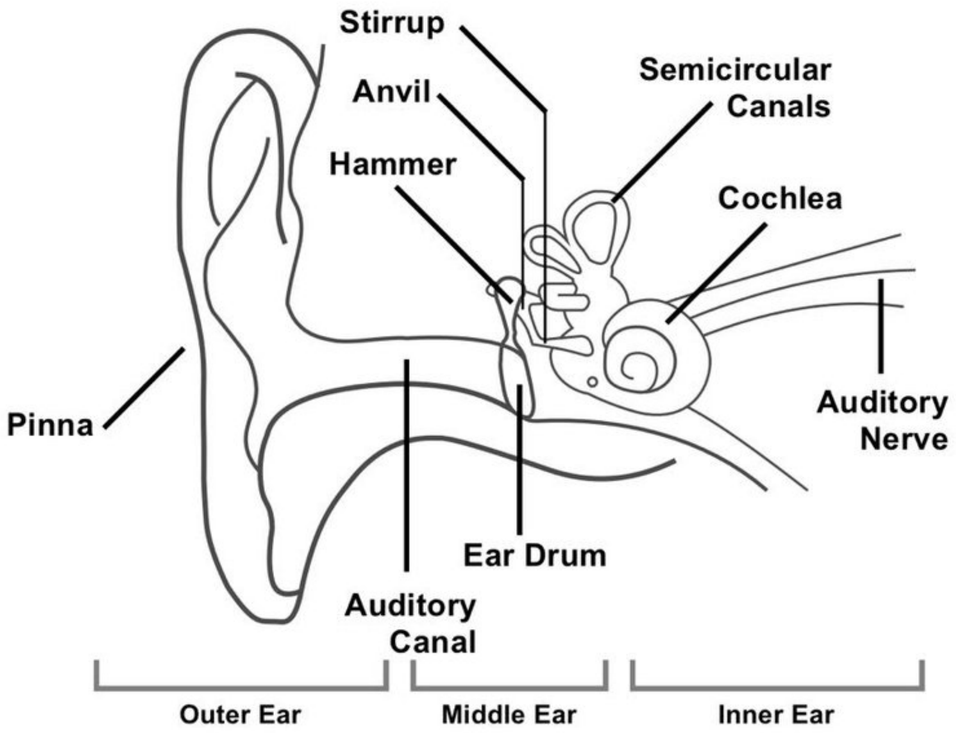
- Human hearing is roughly: 20Hz - 20kHz



From Gorman, Benjamin. (2018). A Framework for Speechreading Acquisition Tools.

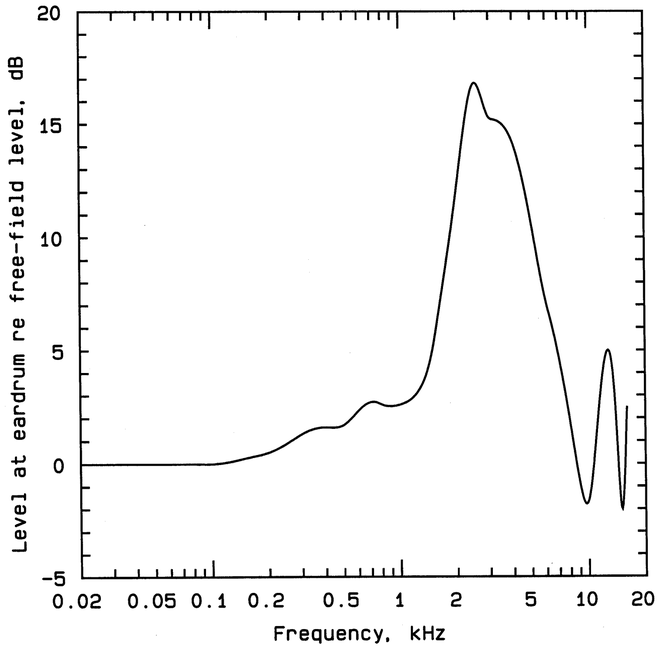
Outer Ear

- Sound waves are reflected into the auditory canal by the *pinna*.
- The *auditory canal* acts as an amplifier for certain frequencies (human voices).
- The *ear drum* vibrates, transferring the sound into the middle ear.

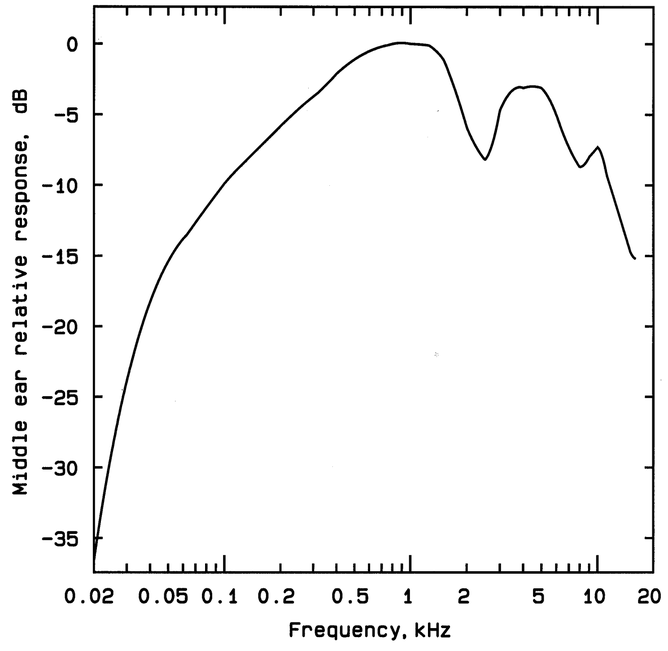


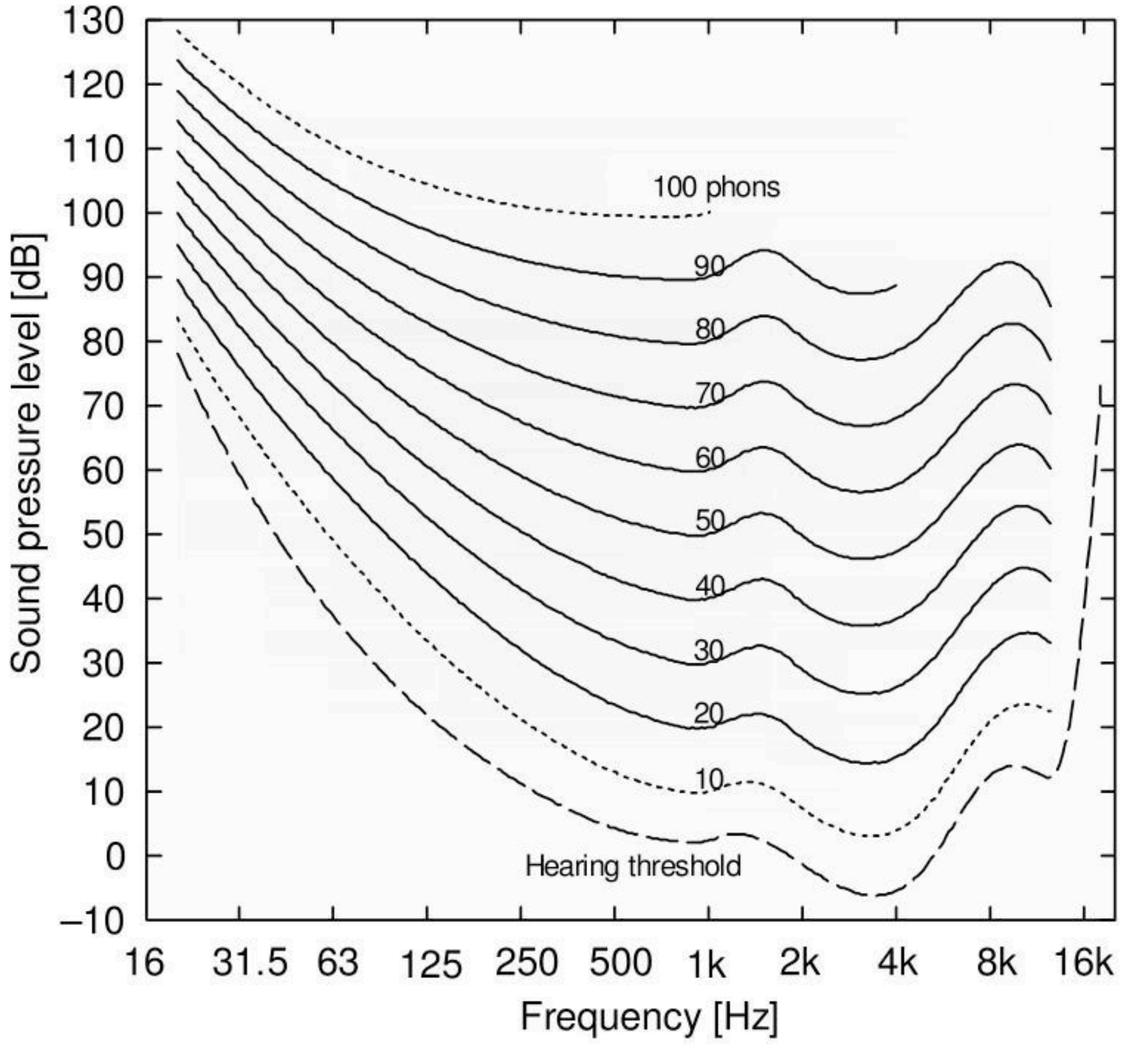
Sum = Equal Loudness Curves

Auditory canal amplification



Ear drum response

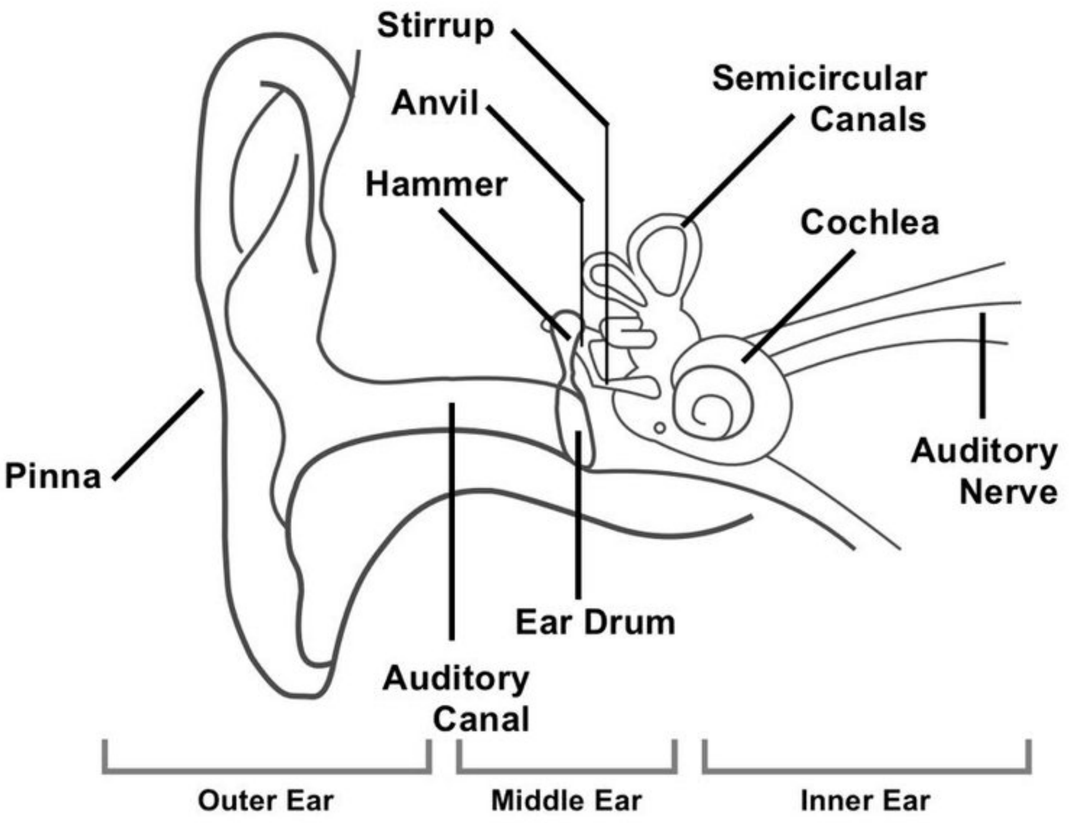




All from Moore, B.C.J. An introduction to the psychology of hearing. 2012.

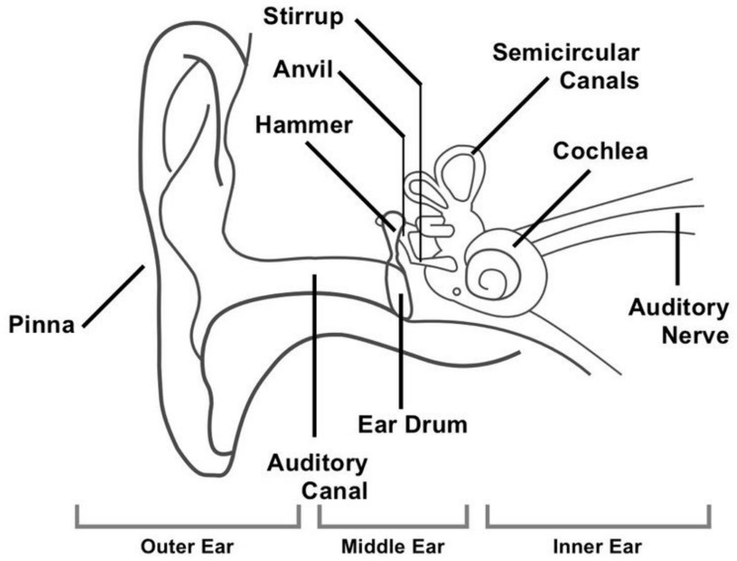
Middle Ear

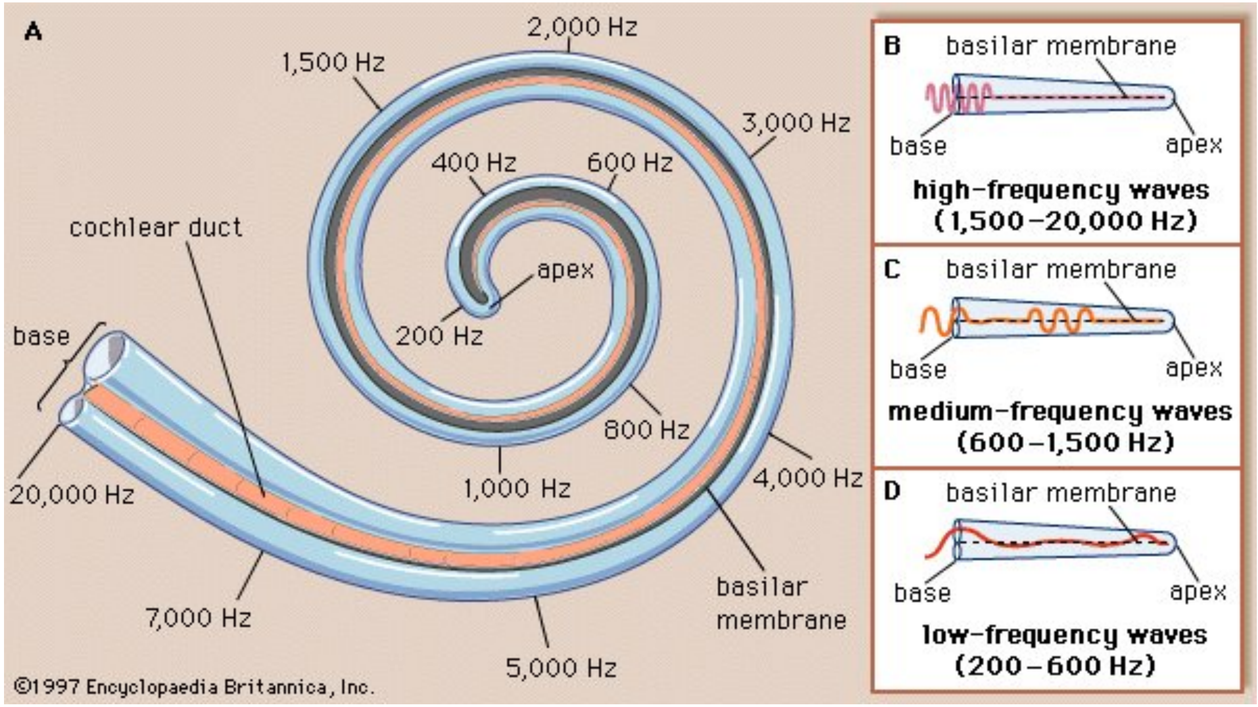
- Ensures the efficient transfer of pressure waves from air outside the ear to higher pressure fluid in the inner ear.



Inner Ear (Cochlea)

- Spiral shape so that it can be longer.
- *Stereocilia* (hairs) along its length respond to different frequencies.
- These responses are sent to the brain as electric signals.





In the brain

- The ear detects *frequencies*
- The brain interprets frequencies as speech, music, etc.

More about the Brain

- The brain can amplify or dampen certain frequencies, for example to hear more clearly in a crowded room.

- It also uses visual information!
  - https://www.youtube.com/watch?v=G-lN8vWm3m0

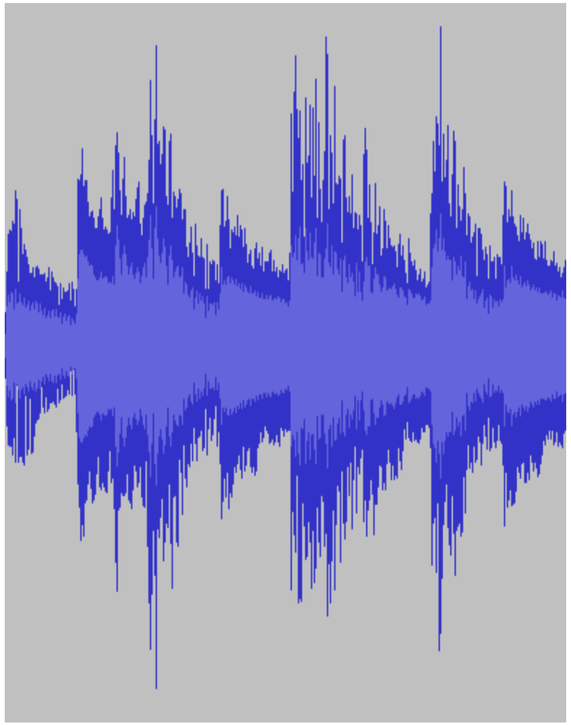
Conclusion

- Sound travels through the air as longitudinal pressure waves.
- The ear's design allows it to:
  - Amplify certain (musical, speech-related) frequencies, and
  - Resonate in response to *frequency* presence in the signal

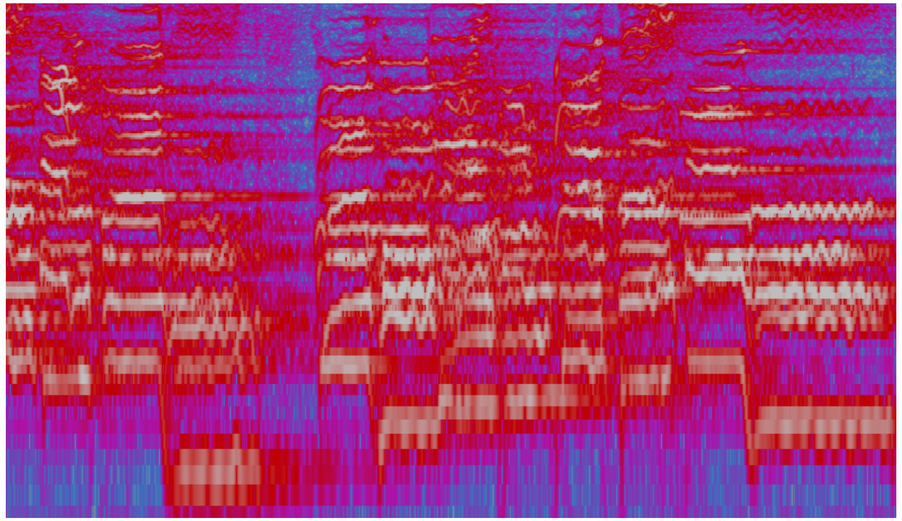
Sound Processing in the Computer



Part 1

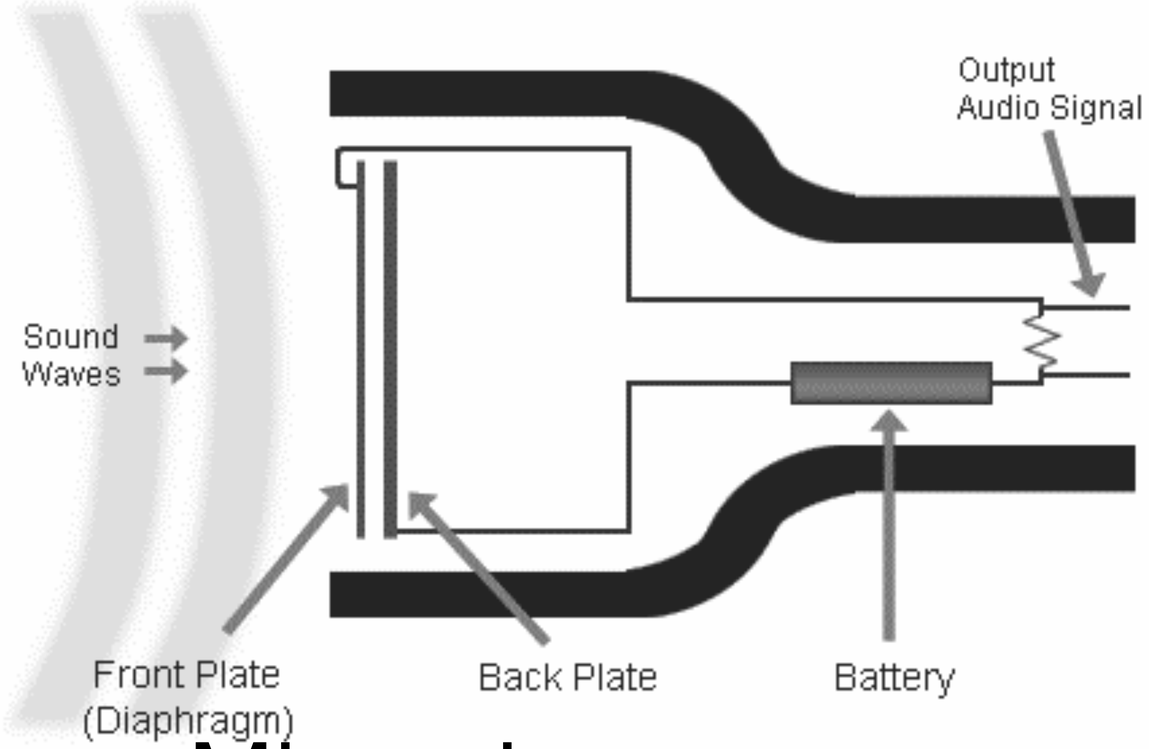


Part 2



Part 1: How is audio stored on a computer?







Microphone

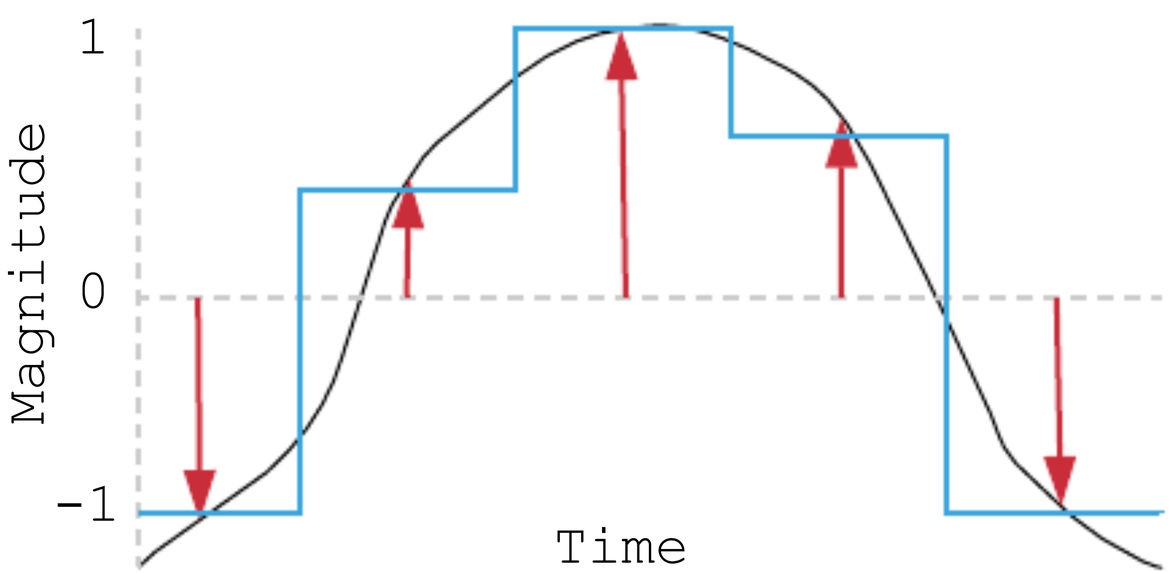
Wave file

```
[0.        ,  0.84147098,  0.90929743,  0.14112001, -0.7568025 , -0.95892427,
-0.2794155 ,  0.6569866 ,  0.98935825,  0.41211849, -0.54402111, -0.99999021,
-0.53657292,  0.42016704,  0.99060736, 0.65028784, -0.28790332, -0.96139749,
-0.75098725,  0.14987721, ...]
```

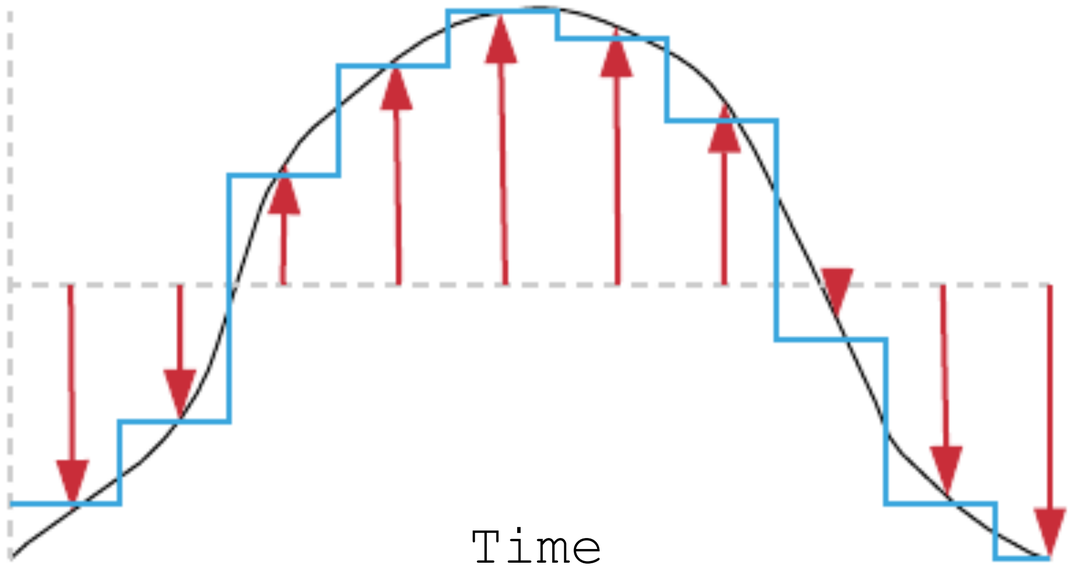
- Each number represents the air pressure detected by the microphone at a given time (-1 ≤ value ≤ 1).

- The *sample rate* of a wave file is the number of these values per second. Commonly 44100 Hz.

Importance of sample rate

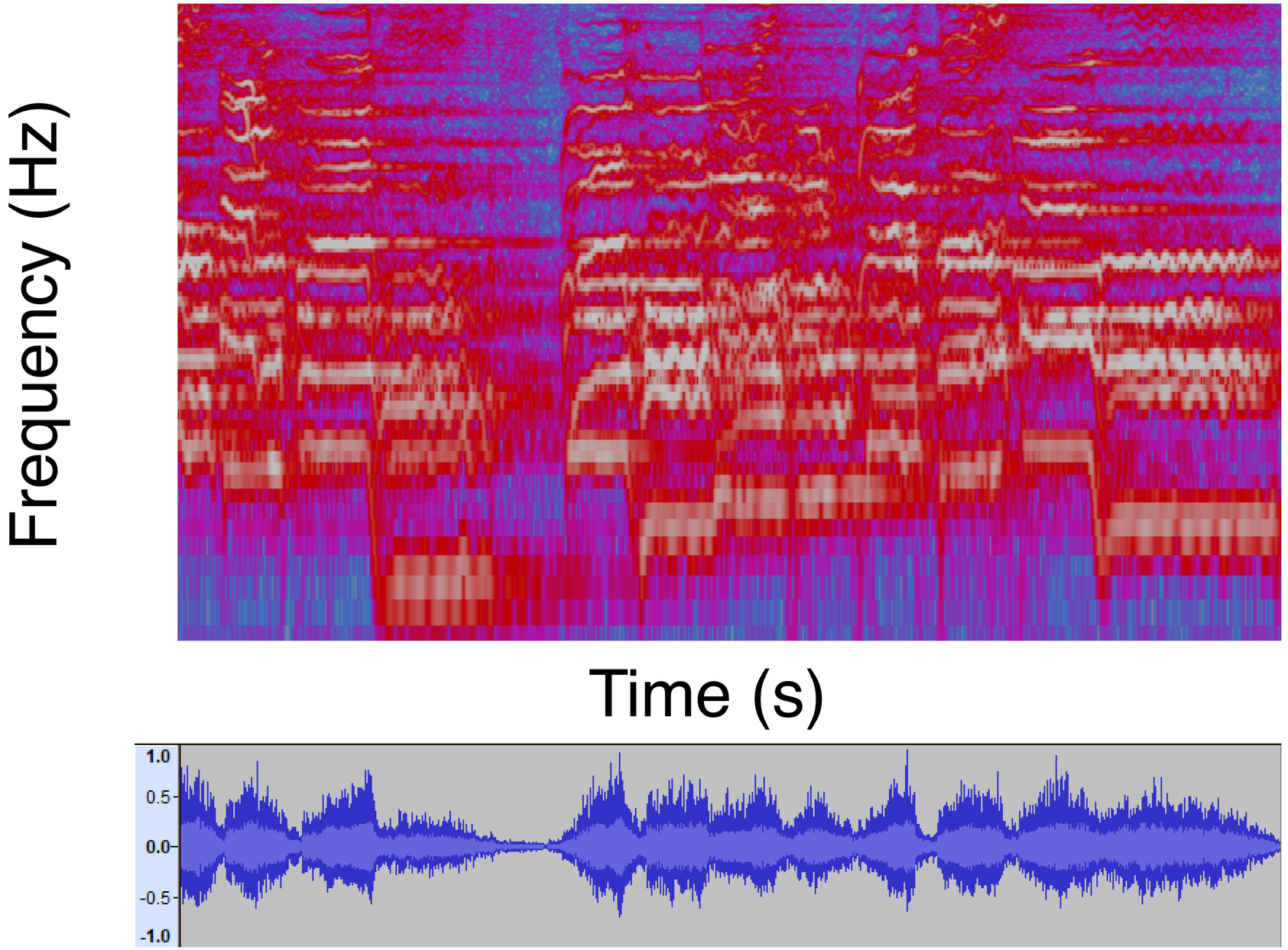


Low sample rate



Higher sample rate

The Spectrogram



Discrete Fourier Transform

- https://www.youtube.com/watch?v=spUNpyF58BY

- More information
  - https://www.youtube.com/watch?v=ds0cmAV-Yek

DFT in practice

- Cut your signal into short, overlapping segments, and compute the DFT of each segment.

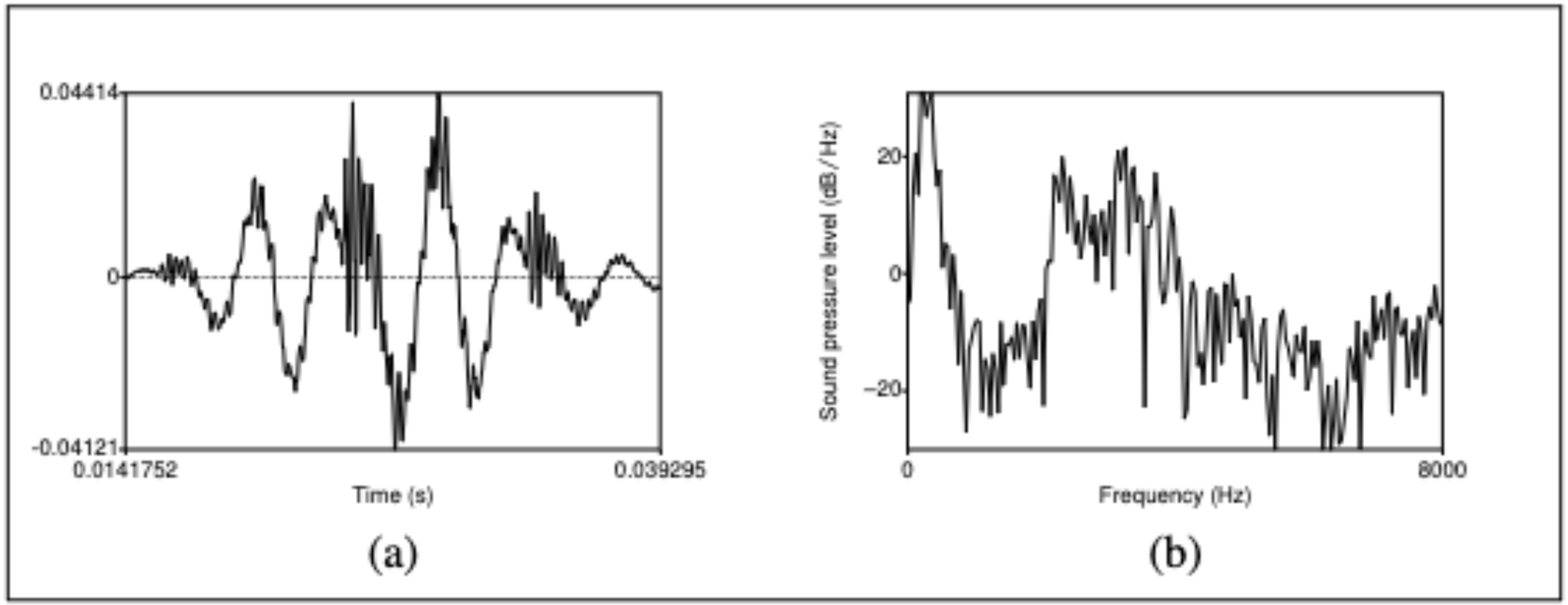


Figure 15.28 (a) A 25 ms Hamming-windowed portion of a signal from the vowel [iy] and (b) its spectrum computed by a DFT.

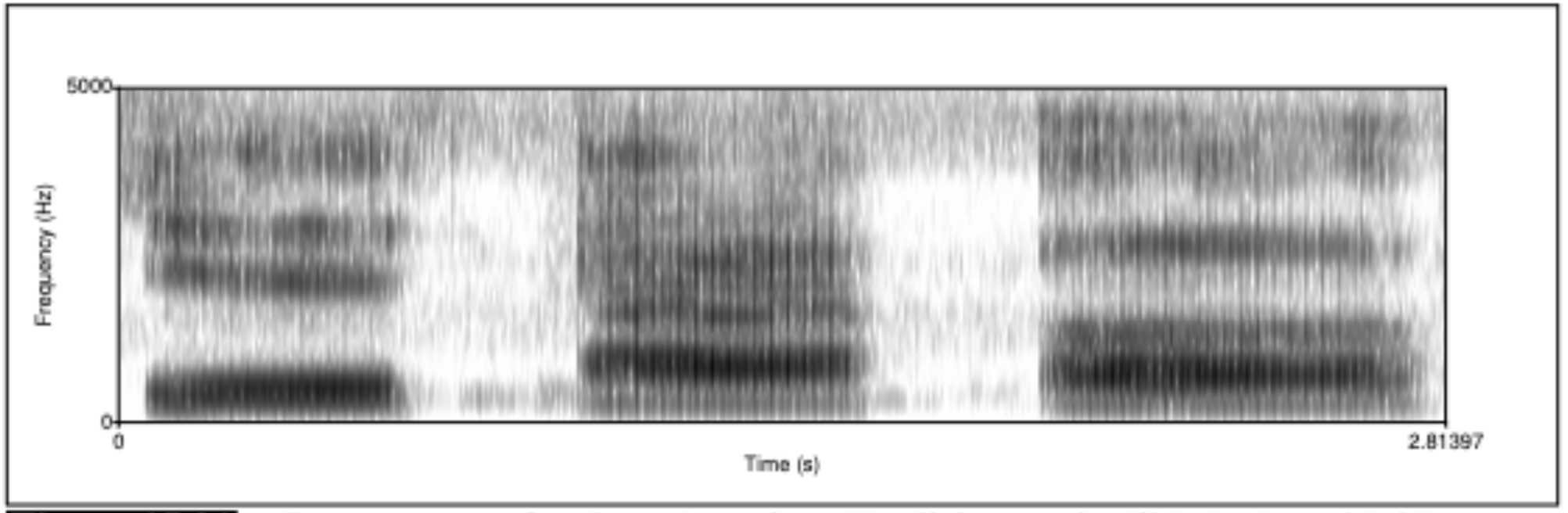


Figure 15.21 Spectrograms for three American English vowels, [ih], [ae], and [uh]

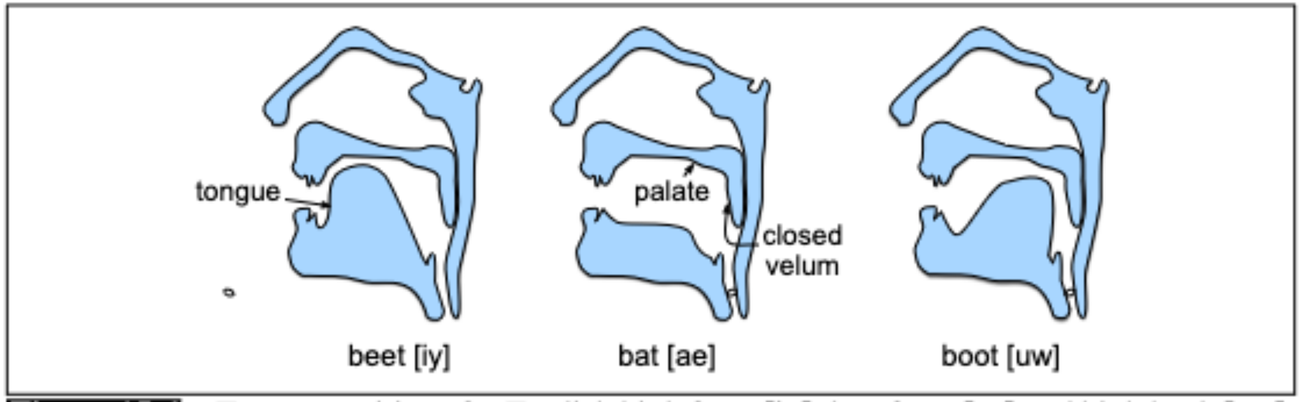


Figure 15.6 Tongue positions for English high front [iy], low front [ae] and high back [uw].

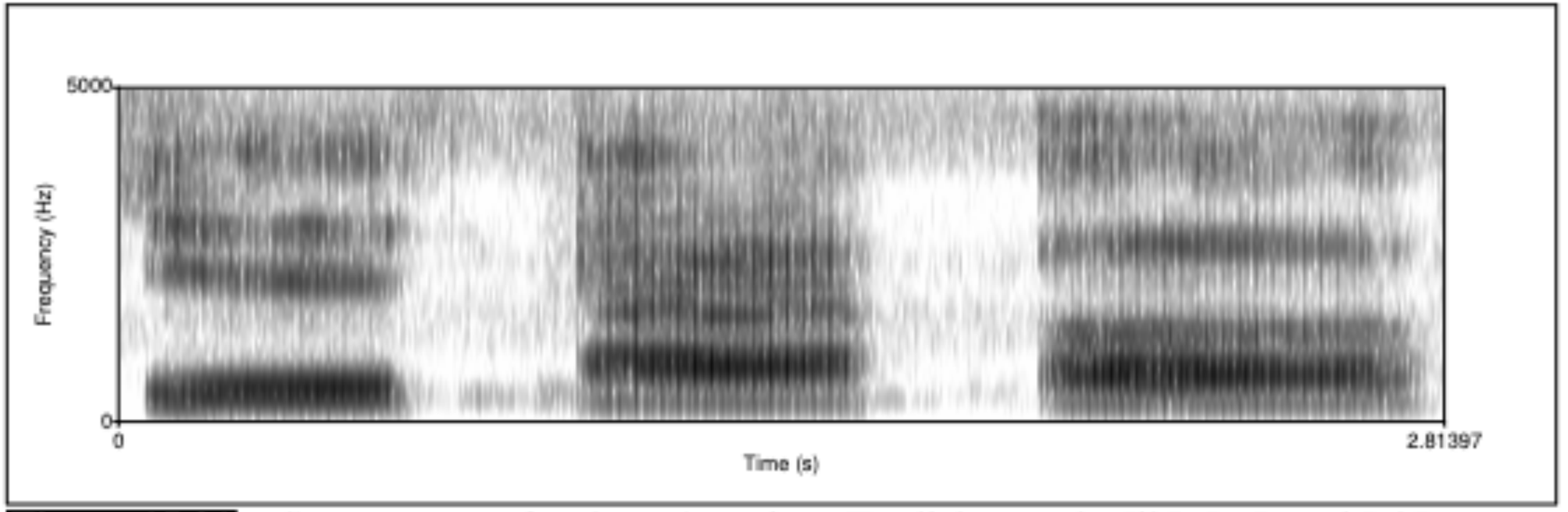


Figure 15.21 Spectrograms for three American English vowels, [ih], [ae], and [uh]

Formants

- Dark band in the spectrogram
- F1 and F2 determine vowel identity
- F3 reflects properties of an individual speaker’s voice tract

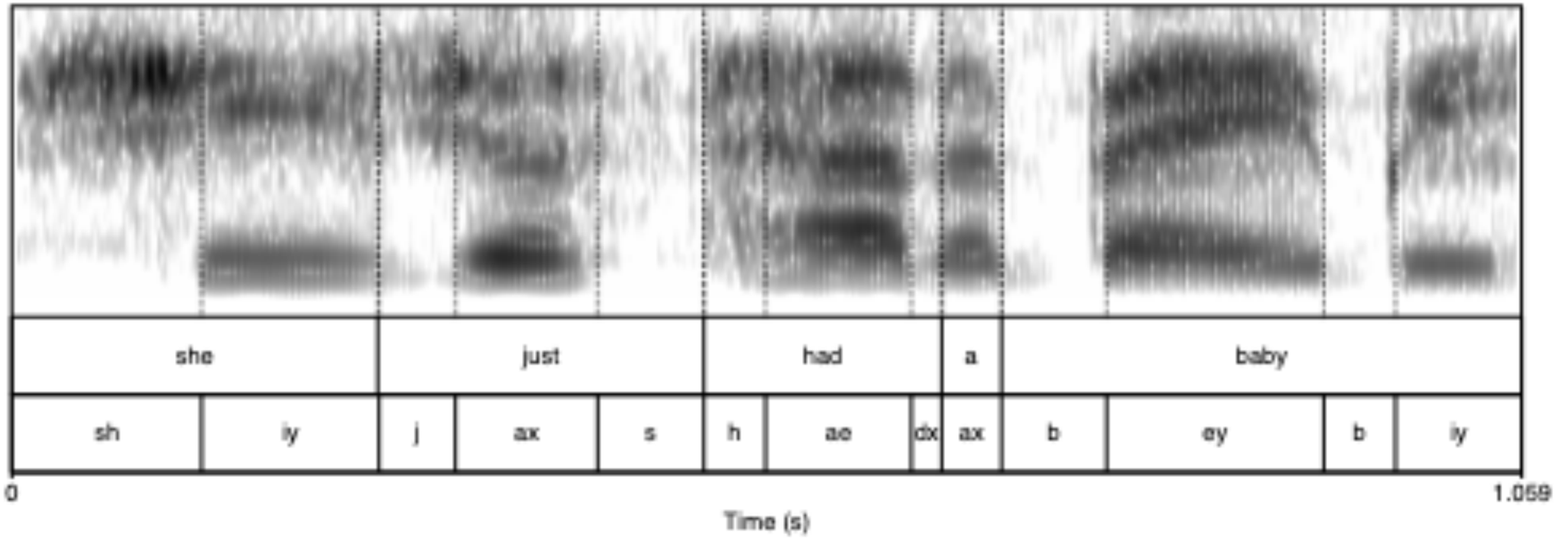


Figure 15.22 A spectrogram of the sentence "she just had a baby" whose waveform was shown in Fig. 15.15. We can think of a spectrogram as a collection of spectra (time slices), like Fig. 15.20 placed end to end.

Signal Processing - Conclusion

- If you want to work with audio data, use a spectrogram!

- Spectrograms are also useful for examining audio recordings manually, so it’s good to understand their structure.